CASE STUDY · STRATEGIC INTELLIGENCE SERIES

Quantitative Market Entry Simulation Engine

A Stochastic Decision-Support Framework for Global Expansion

GE-McKinsey Nine-Box Integration · World Bank API · ETL Automation

Confidential · Internal Strategy Document · Not for Distribution

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Executive Summary

Most market entry decisions get made with a mix of gut feel and outdated reports. Decision-makers drown in contradictory analyst data and macroeconomic indicators stripped of any strategic context. I built this engine to fix that.

It pulls live economic data from the World Bank, runs it through a weighted scoring model, and ranks which markets are actually worth pursuing. The output is a ranked dashboard with one clear takeaway per market: go here, watch this one, or avoid that.

> *"The goal was simple: reduce the time between raw data and a defensible recommendation from days of spreadsheet work to effectively zero."*

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The Problem Statement

Conventional market analysis fails in two predictable ways:

- **Static Bias.** Traditional models treat growth as a straight line. They ignore inflation velocity, currency risk, and the fact that emerging-market growth rates are structurally unsustainable at their early peak. The forecast looks rigorous on paper but collapses against real-world variance.
- **Operational Lag.** Manual data retrieval takes days. Pulling figures from the World Bank portal, normalizing units, reconciling vintages. By the time a report reaches a stakeholder, the underlying figures are already stale.

The engine was built to eliminate both: replace static assumptions with probabilistic modeling, and replace manual pipelines with fully automated ETL.

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Methodology & Design

The scoring model rests on three design choices, each made to address a specific flaw in conventional analysis.

Risk-Adjusted ROI Core

I score each market on four weighted factors: GDP growth (40%), labor capacity (30%), infrastructure quality (20%), and inflation volatility (10%). Inflation is treated as a penalty, not just another input. Markets that grow fast but inflate faster get marked down accordingly.

Maturity Decay Modeling

Emerging-market growth rates are not sustainable at their early-stage peak. The model applies a decay coefficient of 0.95 to terminal value projections. Without this, the engine would over-weight markets that are already past their inflection point and heading toward saturation.

Stochastic Sensitivity Layer

Deterministic models produce identical projections from identical inputs. That false precision breeds overconfidence. I inject calibrated Gaussian noise (sigma = 0.005) into all GDP forecasts. It forces the model to differentiate between superficially similar markets and acknowledges the unpredictability that any real capital deployment has to absorb.

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Strategic Framework: The Nine-Box Matrix

Scored markets are mapped onto the GE-McKinsey Nine-Box Matrix, a framework originally built for multi-business portfolio prioritization and adapted here for cross-border market selection. Each cell drives a clear capital stance.

- **Invest / Grow (Primary Targets).** High-momentum markets with solid infrastructure scores and contained inflation risk. These are the engine's highest-conviction calls.
- **Selectivity / Watch (Tactical Positions).** Markets with moderate volatility or mixed signals. Viable under the right entry vehicle — joint ventures, licensing, or phased FDI — but they need continuous monitoring.
- **Harvest / Divest (Screen Out).** High-inflation, low-infrastructure environments that exceed the model's risk ceiling. Excluded from active screening so analyst bandwidth stays focused on better opportunities.

	Low Attractiveness	Medium Attractiveness	High Attractiveness
Strong Position	Selectivity	Selectivity / Growth	Invest / Grow
Medium Position	Harvest / Divest	Selectivity	Selectivity / Growth
Weak Position	Harvest / Divest	Harvest / Divest	Selectivity

GE-McKinsey Nine-Box · Entry Risk Classification

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Technical Implementation

Every engineering decision came back to one constraint: the dashboard has to be current, fast, and self-maintaining without any manual work on my end.

Automated ETL Pipeline

A GitHub Actions workflow runs on a weekly schedule. It queries the World Bank API, normalizes the response payloads, and writes a validated snapshot to market_engine_cache.csv. If the API returns stale or malformed data, the pipeline raises a build failure rather than silently updating the cache with bad figures.

Cache-First Architecture

Early versions of the dashboard called the World Bank API directly on each page load. That caused two problems: slow load times and brittleness whenever the API was unavailable. The cache decouples the two completely. The frontend reads from a pre-processed CSV, which keeps load times under a second regardless of what the upstream API is doing.

Interactive Visual Layer

The dashboard has two deliberate views: a macro-level Nine-Box scatter plot for portfolio-level prioritization, and a country drill-down exposing the raw metric distributions. The goal was to let someone pivot between the strategic overview and the underlying data without leaving the interface.

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Strategic Impact

The engine reframes market entry analysis as an engineering problem. Its impact operates across three dimensions.

- **Speed.** Information-to-insight latency drops from days of manual work to effectively zero. The moment World Bank data updates, every market in the registry gets re-scored automatically.
- **Consistency.** Every market gets evaluated against the same quantitative framework. The model removes analyst subjectivity and recency bias, two of the most common sources of misjudgment in market selection.
- **Scalability.** Adding a new country requires no code changes. Just a World Bank country code. The architecture can expand from dozens of markets to hundreds without touching the pipeline.

> *"By standardizing the analytical baseline, the engine does not replace strategic judgment. It clears the noise so that judgment can operate at a higher level."*